0191. Number of 1 Bits

sy | Category: Bit Manipulation | LeetCode: https://leetcode.com/problems/numb

Problem

- Return the number of set bits (Hamming weight) of an integer treated as unsigned 32-bit.

Intuition

- N & (n - 1) clears the lowest set bit. Count how many times we can do that before n is 0.

Brute Force

- Check all 32 bits with (n >>> i) & 1.

Optimal Approach

- Count = 0; while n != 0: n &= n - 1; count++.

Worked Example

```
n = 11 (1011).

1011 & 1010 = 1010  count 1
1010 & 1001 = 1000  count 2
1000 & 0111 = 0000  count 3
```

Java Solution

```java
public class NumberOf1Bits {
    public int hammingWeight(int n) {
        int count = 0;
        while (n != 0) { n &= n - 1; count++; }
        return count;
    }
}
```

Complexity

- Time: O(number of set bits). Space: O(1).

Pitfalls

- Use n != 0, not n > 0: negative ints have the sign bit set.